

ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Optional Preliminary Full Draft

Purpose

This optional submission gives you an opportunity to receive feedback on a complete draft of your term project before the final paper is due. The draft should represent your best current version of the entire paper. It does not need to be fully polished, but all major sections, analyses, figures, and interpretations should be present. This submission does not replace the final paper.

Required Content

Title and abstract

Provide a descriptive title and a brief abstract summarizing the question, methods, primary results, and main interpretation.

Introduction

Explain:

- The ecological or conservation problem
- Relevant background
- The research question
- Your expectations or hypotheses

Methods

Describe the complete modeling workflow:

- Occurrence-data sources and filtering
- Definition and biological justification of M
- Background sampling
- Environmental predictors and data processing
- Model algorithm and complexity settings
- Model evaluation
- Robustness or sensitivity analyses
- Projection procedures, if applicable

Include enough detail for another researcher to understand and reproduce the analysis.

Results

Present the primary results without extensive interpretation. Include:

- Data and sample-size summaries
- Model-evaluation results
- Relative-suitability maps or other primary predictions
- Response curves or variable-importance results, when relevant
- Comparisons across alternative assumptions

All figures and tables should have numbers, descriptive captions, and references in the text.

Discussion

Interpret the results in relation to the research question and biological expectations. Discuss:

- The biological meaning of the modeled patterns
- Whether conclusions were stable across assumptions
- Important limitations
- Alternative explanations
- Appropriate and inappropriate uses of the model
- Future data or analyses that would improve the study

Reproducibility

Confirm that the scripts, model settings, figures, and documentation in your repository correspond to the submitted draft. Include the private GitHub repository link.

References

Include a separate References section and use a consistent citation format. Cite relevant sources for:

- The ecology and natural history of the study organism
- Biological mechanisms connecting the predictors to the species
- The geographic or conservation context of the project
- Occurrence and environmental datasets
- Modeling and evaluation methods
- Software or specialized R packages

Ecological statements about habitat, physiology, dispersal, species interactions, or conservation concerns should be supported by relevant literature. Use peer-reviewed primary research when possible. Include in-text citations as you typically read in journal articles and complete entries in the References section.

What makes a draft ready for feedback?

The draft should contain complete sections. Minor placeholders are acceptable, but missing analyses or figures will limit the feedback you receive. Feedback will prioritize:

- Alignment between the research question and analysis
- Modeling assumptions and evaluation
- Biological interpretation

- Organization and clarity
- Figures, captions, and evidence
- Remaining work needed for the final submission

Format and submission

- Optional submission, due Tuesday, December 1 by 11:59 PM
- Submit as a PDF or Word document through Canvas
- Use 12-point Times New Roman and standard margins
- Include your private GitHub repository link
- Final paper due Friday, December 11

This submission is intended for formative feedback. You remain responsible for revising, checking, and completing the final paper.